Chapter 1 exercises

State the set being counted, the counting rule, and the calculation.

DM01-E01

A school course code has three distinct uppercase English letters followed by two distinct digits. Neither O nor I is permitted, and the first digit cannot be zero. How many codes exist?

DM01-E02

Choose a chair, a vice chair, and three committee members from eight candidates. Each person has at most one role, and the three ordinary members have no ordered offices. How many assignments record these roles and identities?

DM01-E03

How many arrangements of all letters of BANANARAMA begin and end with A?

DM01-E04

Give a combinatorial proof of

$$\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1}, \quad n \geq 1.$$

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https://github.com/xinhi6716/computer-science-self-study

DM01-E05

How many nonnegative integer solutions of $x_1 + x_2 + x_3 + x_4 = 13$ have every entry at least two?

DM01-E06

An admissible word has n opening and n closing parentheses, with no prefix having more closings than openings. Use Catalan numbers to count these words for $n = 6$, and explain why $\binom{12}{6}/2$ is invalid.

DM01-E07

Count arrangements of BANANARAMA with no adjacent A's. Then also require A at both ends. Explain why the gap encoding is uniquely reversible.

DM01-E08

A nine-bit word has six 1's and three 0's. Count words with exactly four runs and with exactly five runs. If the number of 0's becomes zero while six 1's remain, can the two-positive-count formula still be used directly?

DM01-E09

Put seven distinct books on three labeled shelves with left-to-right order. Count placements with no empty shelf and placements allowing empty shelves. Compare with recording only each book's destination, and explain why the three answers do not conflict.

DM01-E10

Find $[x^2y^3z^2](x-2y+z)^7$ and the sum of all coefficients. How should the factor-selection argument be interpreted when coefficients may be negative?

DM01-E11

A length-eleven step word contains seven $+1$'s and four -1's. The prefix sum is the sum of all step values in that prefix. Count words for which every prefix sum is nonnegative. State the reflection target for inadmissible words and its inverse explicitly.

DM01-E12

Count ordered full binary trees with five internal nodes, triangulations of a convex heptagon, and full parenthesizations of six factors in fixed order. Explain their Catalan parameters. What changes for seven factors?

DM01-E13

Let n be a nonnegative integer. Count integer triples $1 \le k < j \le i \le n$, using a bijection to strictly increasing subsets, and evaluate at $n = 8$. How does this differ from the lecture's loop with both inequalities weak?

DM01-E14

Count integer solutions of $x_1 + x_2 + x_3 + x_4 = 11$, with $x_1 \geq 2, x_2 \geq 1, x_3 \geq 0, 0 \leq x_4 \leq 3$. Remove lower bounds first, then use a complement for the single upper bound. Do not directly multiply purported fixed choices for the entries.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 1.

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